



Carisurg

Assignment 17

Final Documentation

Due Date: July 14th, 2026.

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Setup & Data Decisions

This baseline was built on 55,121 emergency department encounters from `triage_cleaned_v1.csv`, the canonical output of the Week 5 cleaning pipeline. A random seed of 42 was committed to the README to ensure reproducibility. The data was split 80/20 into a training set of 44,096 patients and a test set of 11,025, with stratification on the ESI column to preserve the class distribution in both halves.

Two columns were excluded on leakage grounds: `disposition` and `previousdispo`. Both are only known after the encounter ends, and including them would inflate scores while producing a model that collapses in real clinical use. The arrival mode column was encoded as a binary ambulance flag, noting that the value 'ambulance' is lowercase in this dataset, accounting for 18,561 arrivals or 33.7% of the total.

Class imbalance was addressed by applying `class_weight='balanced'` to both classifiers. ESI Level 1 represents only 77 patients, or 0.14% of the dataset. Without balancing, the model learns to ignore ESI 1 almost entirely, since predicting it incorrectly has little effect on overall accuracy. The ten features used were drawn from the Week 5 memo shortlist: age, SpO₂, chest pain, shortness of breath, suicidal presentation, alcohol intoxication, altered mental status, ambulance arrival, respiratory rate, and heart rate.

Primary Metric Justification

The primary metric is recall for ESI Level 1, with macro-averaged recall across all five classes used as the summary metric.

ESI 1 patients — cardiac arrest, major trauma, anaphylaxis, shock — require an immediate resuscitation bay response. A false negative on ESI 1 does not mean a slightly delayed response: in cardiac arrest, survival probability drops roughly 10% per minute without intervention. A model that classifies a Level 1 patient as Level 2 or 3 sends them into a waiting queue. The queue is the wrong place for them.

In this dataset, ESI 1 is 0.14% of encounters, representing 77 patients in total and 16 in the test set. Overall accuracy is therefore clinically meaningless: a model that never predicts ESI 1 at all achieves 99.86% accuracy while failing every Level 1 patient it sees. Recall asks the right clinical question — of all patients who were actually ESI 1, how many did the model correctly flag?

Weighted F1 is dominated by ESI 3, which makes up 49% of the data. A model that ignores ESI 1 entirely can still score well on weighted F1, which makes it the wrong optimisation target for a triage tool. Macro F1 gives equal weight to every class, ensuring that poor ESI-1 performance is visible rather than buried under majority-class results. Both are reported, but macro recall on ESI 1 is the number that matters most. The failure mode we are most worried about is false negatives on ESI 1 — a cardiac arrest patient classified as ESI 2 or 3, sent to wait in a general acute cubicle.

Results

Logistic regression is the stronger baseline on every metric that matters. It achieved an ESI-1 recall of 0.750, meaning it caught 12 of the 16 ESI-1 patients in the test set. Random guessing would catch approximately 2. Both models beat the dummy on macro recall, confirming they extract genuine signal from the features rather than simply memorising the class distribution.

Looking at per-class performance for logistic regression, ESI-2 recall was 0.31, which is a notable concern since ESI-2 patients — stroke, sepsis, ACS — have a 10-minute physician target. This is a direct trade-off from `class_weight='balanced'`, which up-weights rare classes at the cost of majority-class precision. The decision tree caught 9 of 16 ESI-1 patients for a recall of 0.563, but assigned ESI 4 to zero patients entirely. This is a known limitation of shallow decision trees under balanced weighting on highly imbalanced data — at depth 5, the tree cannot find a split that improves on the weighted Gini impurity for ESI 4. A deeper tree or a different weighting scheme would be needed to address this.

Model	Accuracy	Macro Recall	ESI-1 Recall	Weighted F1	Macro F1
Dummy (stratified)	0.375	0.204	~0.14	0.375	0.204
Logistic Regression	0.254	0.417	0.750	0.327	0.214
Decision Tree (depth=5)	0.255	0.386	0.563	0.302	0.170

Table 1. Head-to-Head Comparison

Class	Precision	Recall	F1-Score	Support
ESI 1	0.00	0.75	0.01	16
ESI 2	0.69	0.31	0.42	3,585
ESI 3	0.58	0.20	0.30	5,402
ESI 4	0.25	0.26	0.26	1,779
ESI 5	0.05	0.57	0.09	243

Table 2. Logistic Regression Per-Class Results

Class	Precision	Recall	F1-Score	Support
ESI 1	0.01	0.56	0.01	16
ESI 2	0.50	0.40	0.44	3,585
ESI 3	0.57	0.22	0.32	5,402
ESI 4	0.00	0.00	0.00	1,779
ESI 5	0.04	0.75	0.08	243

Table 3. Decision Tree Per-Class Results

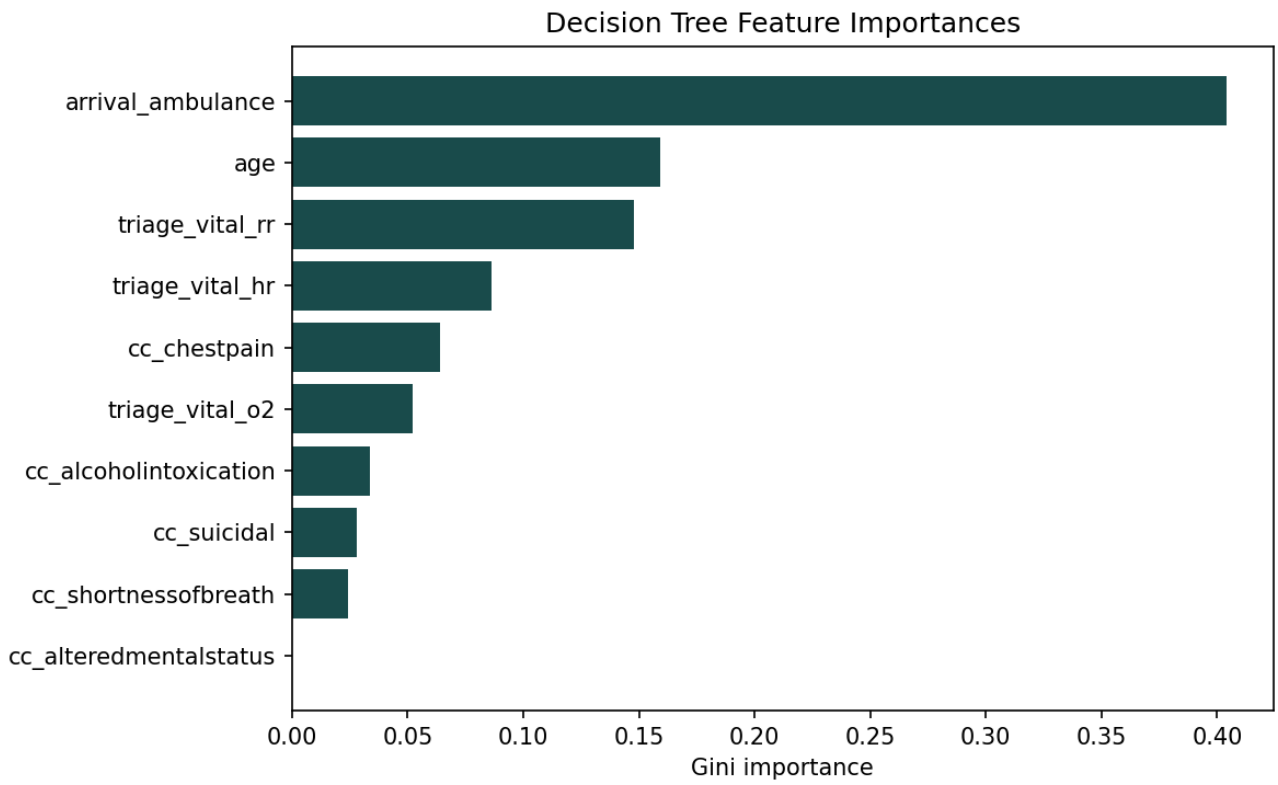


Figure 1. Decision Tree Feature Importances

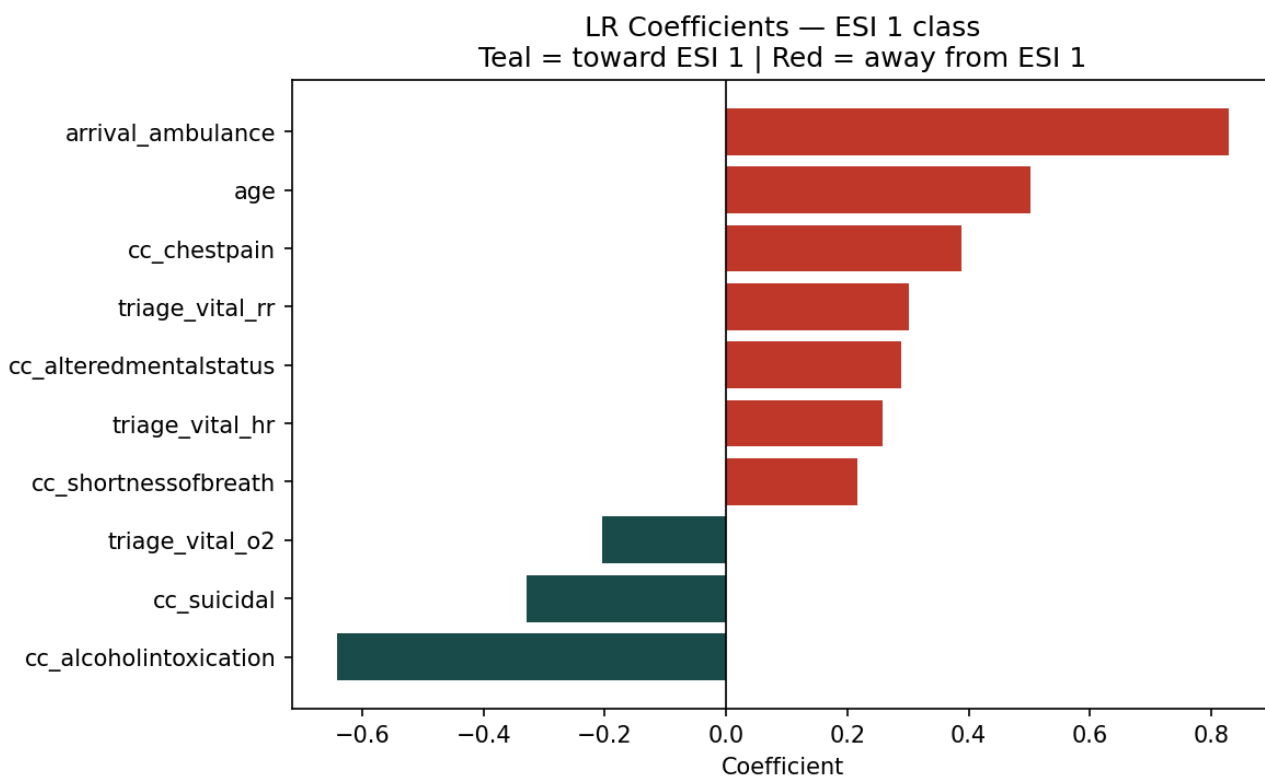


Figure 2. Logistic Regression Coefficients for the ESI 1 class (Teal = toward ESI 1, red = away from ESI 2)

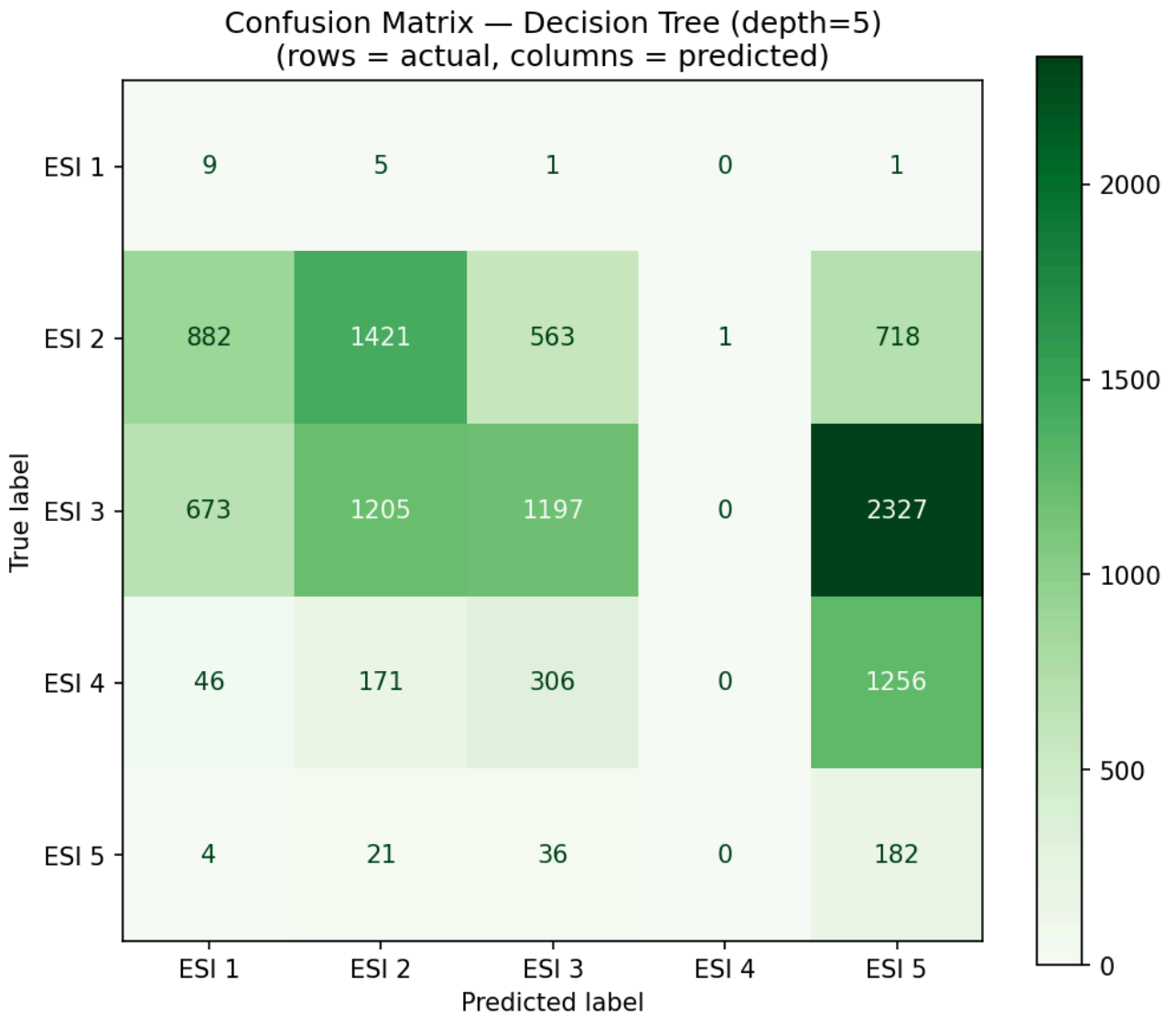


Figure 3. Confusion Matrix—Decision Tree (Depth = 5), (Rows = Actual, Columns = Predicted.)

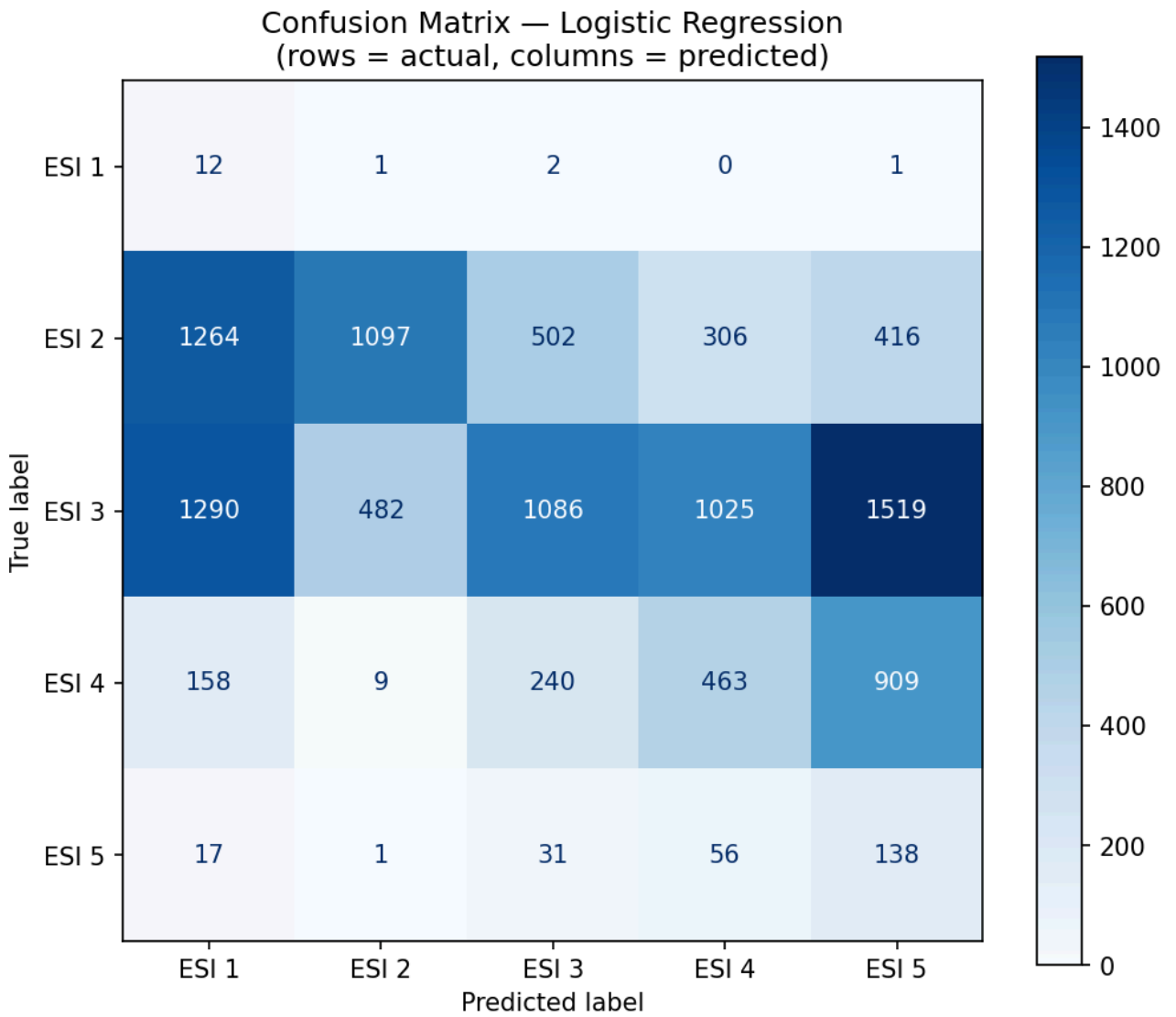


Figure 4, *Confusion Matrix — Logistic Regression. (Rows = Actual, Columns = Predicted)*

Failure Mode Reflection

The primary concern is that logistic regression misses 4 of 16 ESI-1 patients, a false negative rate of 25%. In a deployed context, those patients would join a waiting queue rather than go to the resuscitation bay. For a cardiac arrest patient, this outcome is unsurvivable.

The ESI-2 recall failure is also worth naming directly. A recall of 0.31 for ESI 2 means the majority of stroke, sepsis, and ACS patients are being missed at their urgency level. This trade-off requires threshold calibration before any deployment is considered. Additionally, a static classifier has no re-triage capability — a patient correctly classified as ESI 3 at arrival who deteriorates to ESI 1 within 30 minutes is not re-flagged. The tool must complement the nurse's visual scanning of the waiting room, not replace it.

On the ESI-1 recall figure itself, caution is warranted. With only 16 ESI-1 patients in the test set, each individual prediction shifts recall by 6.25 percentage points. The 0.750 figure is directionally encouraging but statistically fragile and should not be over-interpreted. The overriding caveat on everything above is the Caribbean mismatch: this is a US, single-hospital dataset. Nothing here tests whether these models would transfer to Mercer General. Phase 1 — building Caribbean-sourced data — remains the hard prerequisite for any deployment-oriented claim.

Appendices

Supplementary Links

1. The Loom link can be found [here](#).
2. The GitHub link can be found [here](#)
3. Evidence for every figure above is included alongside this memo in the GitHub:
 - a. **figs/01-06_*.png**
 - b. **notebooks/Week6_Baseline_Model**

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol